

Recurrence relation:-

Principle of Inclusion and Exclusion

* First order Recurrence relation:-

A first order Recurrence relation with constant coefficient is of the form

$$a_n = c a_{n-1} + f(n) \quad \text{--- (1) for } n \geq 1$$

where c is a known constant

$f(n)$ is a known function of n .

If $f(n) = 0$, then eqn (1) is called a homogeneous recurrence relation otherwise it is called non-homogeneous recurrence relation.

The general solution of a homogeneous recurrence relation

$$\text{is } a_n = c^n a_0 \quad \text{for } n \geq 1.$$

The general solution of a non-homogeneous recurrence

$$\text{relation of order-1 is given by } a_n = c^n a_0 + \sum_{k=1}^n c^{n-k} f(k)$$

for $n \geq 1$.

0, 2, 4, 6, 8, 10, -

General term

$$a_n = 2n$$

Recurrence relation

$$a_n = a_{n-1} + 2, \quad a_0 = 0, a_1 = 2$$

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17 Solve the recurrence relation $a_n = 7a_{n-1}$, where $n \geq 1$
given $a_2 = 98$.

Sol

$$a_n = 7a_{n-1}, \quad a_2 = 98.$$

General solution is ~~the~~ homogeneous 1st order

$$a_n = c^n a_0.$$

$$a_n = 7^n a_0. \quad \text{--- (2)}$$

$$\text{for } n=2, \quad a_2 = 7^2 a_0 \Rightarrow a_2 = 49 a_0.$$

$$98 = 49 a_0$$

$$\boxed{a_0 = 2}$$

Sub. in (2),

$$\boxed{a_n = 2 \cdot 7^n.} \quad \text{General solution.}$$

26 Solve the recurrence relation $a_n = n a_{n-1}$ where $n \geq 1$ and
7-0-2
7(1)
 $a_0 = 1$.

Sol Given $a_n = n a_{n-1}$ --- (1)

homogeneous 1st order

$$n=1, \quad a_1 = 1 \cdot a_0.$$

$$n=2, \quad a_2 = 2 \times a_1 = 2 \times 1 \times a_0.$$

$$n=3, \quad a_3 = 3 \times a_2 = 3 \times 2 \times 1 \times a_0.$$

$\therefore$ General solution is

$$a_n = n! a_0 \quad \text{for } n \geq 1.$$

Using $a_0 = 1$,

$$\boxed{a_n = n!}$$

3) Solve the recurrence relation

$$a_n - 3a_{n-1} = 5 \times 3^n \quad \text{for } n \geq 1 \quad \text{given that } a_0 = 2.$$

Sol.

$$a_n - 3a_{n-1} = 5 \times 3^n \quad \text{--- (1)}$$

— Non-homogeneous recurrence relation,

with $c = 3$, $f(n) = 5 \times 3^n$.

General solution is

$$a_n = c^n a_0 + \sum_{k=1}^n c^{n-k} f(k)$$

$$a_n = 3^n a_0 + \sum_{k=1}^n 3^{n-k} f(k)$$

$$a_n = 3^n a_0 + 3^{n-1} f(1) + 3^{n-2} f(2) + \dots + 3^0 f(n)$$

$$a_n = 2 \times 3^n + 3^{n-1} (5 \times 3^1) + 3^{n-2} (5 \times 3^2) + \dots + 3^0 (5 \times 3^n)$$

$$= 2 \times 3^n + 5 \times \underbrace{[3^n + 3^n + \dots + 3^n]}_{n \text{ times}}$$

$$a_n = 2 \times 3^n + 5 \times n \times 3^n.$$

$$a_n = 3^n (2 + 5n)$$

— Required solution

4/ Find the recurrence relation and initial condition for the sequence 0, 2, 6, 12, 20, 30, 42, ...
Hence find the general term of the sequence.

Sol

Given sequence 0, 2, 6, 12, 20, 30, 42, ...

$$a_0 = 0, a_1 = 2, a_2 = 6, a_3 = 12, a_4 = 20, \dots$$

Consider $a_1 - a_0 = 2 - 0 = 2 = 2 \times 1$

$$a_2 - a_1 = 6 - 2 = 4 = 2 \times 2$$

$$a_3 - a_2 = 12 - 6 = 6 = 2 \times 3$$

$$a_4 - a_3 = 20 - 12 = 8 = 2 \times 4$$

$\vdots$

$a_n - a_{n-1} = 2 \times n$ is the recurrence relation with the initial condition $a_0 = 0$.

Adding all these,

$$a_n - a_0 = (2 \times 1) + (2 \times 2) + (2 \times 3) + \dots + (2 \times (n-1)) + (2 \times n)$$

$$a_n - 0 = 2 [1 + 2 + \dots + n]$$

$$a_n = 2 \left[\frac{n(n+1)}{2} \right]$$

$$\boxed{P_n = n(n+1)}$$

5) The number of virus affected files in a system is 1000 (to start with) and this increases 250% every 2 hours. Use a recurrence relation to determine the no. of virus affected files in the system after one day.

Sol Let $a_0 = 1000$.

Let a_n denote the no. of virus affected files after $2n$ hours.

It increases by 250% every 2 hours.

$$a_1 = a_0 + 250\% a_0$$

$$a_2 = a_1 + 250\% a_1$$

$$a_3 = a_2 + 250\% a_2$$

— — —

$$a_n = a_{n-1} + 250\% a_{n-1}$$

$$a_n = a_{n-1} [1 + 250\%]$$

$$a_n = a_{n-1} \left[1 + \frac{250}{100} \right]$$

$$\boxed{a_n = 3.5 a_{n-1}} \quad \forall n \geq 1$$

This is the recurrence relation for no. of virus affected files
 Its solution is $a_n = C^n a_0$

$$a_n = (3.5)^n \times 1000$$

This gives no. of virus affected files after $2n$ hours

$\therefore$ No. of virus affected files after 24 hours ($n=12$)

$$a_{12} = (3.5)^{12} \times 1000 = 3379220808$$

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Second order homogeneous recurrence relation,

A second order homogeneous recurrence relation is of the form $C_n a_n + C_{n-1} a_{n-1} + C_{n-2} a_{n-2} = 0 \quad \forall n \geq 2$ (1)

where C_n, C_{n-1}, C_{n-2} are real constants

The auxiliary equation of eqn (1) is given by

$$C_n k^2 + C_{n-1} k + C_{n-2} = 0.$$

Suppose k_1 and k_2 are the roots of A.E.

Case (i)

If k_1 and k_2 are real and distinct, then general solution of eqn (1) is given by

$$a_n = A k_1^n + B k_2^n; \text{ where } A \text{ \& } B \text{ are arbitrary constants}$$

Case (ii)

If $k_1 = k_2 = k$, then general soln of eqn (1) is given by $a_n = (A + Bn)k^n$.

Case (iii)

If k_1 and k_2 are imaginary i.e. if $k_1 = p + iq$, and $k_2 = p - iq$, then the general solution of eqn (1) is given by

$$a_n = r^n [A \cos n\theta + B \sin n\theta] \text{ where}$$

$$r = \sqrt{p^2 + q^2}$$

$$\theta = \tan^{-1} \left(\frac{q}{p} \right)$$

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4. Solve the recurrence relation

$$a_n + a_{n-1} - 6a_{n-2} = 0 \quad \text{for } n \geq 2 \text{ given } a_0 = -1, a_1 = 2$$

Sol

$$a_n + a_{n-1} - 6a_{n-2} = 0, \quad a_0 = -1, a_1 = 2$$

(1)

Comparing with

$$C_n a_n + C_{n-1} a_{n-1} + C_{n-2} a_{n-2} = 0,$$

$$C_n = 1, C_{n-1} = 1, C_{n-2} = -6$$

$$\text{A.E. is } C_n k^2 + C_{n-1} k + C_{n-2} = 0,$$

$$k^2 + k - 6 = 0$$

$$(k+3)(k-2) = 0$$

$$\therefore k = -3, 2$$

$k_1 = -3, k_2 = 2$, real & distinct roots,

General solution of (1) is given by

$$a_n = A(-3)^n + B \cdot 2^n \quad \text{--- (2)}$$

Given $a_0 = -1, a_1 = 2$.

Sub $n=0$ in (2),

$$a_0 = A + B$$

$$-1 = A + B \quad \text{--- (3)}$$

Sub $n=1$ in (2),

$$a_1 = A(-3) + B(2)$$

$$2 = -3A + 2B \quad \text{--- (4)}$$

Solving (3) & (4), $A = -2, B = 1$.

Sub in (2), $a_n = (-2)(-3)^n + 1 \cdot (2)^n$

$$a_n = 2^n - 2(-3)^n$$

Q.1) Solve the recurrence relation

M.V. (C) $a_n = 2a_{n-1} - 2a_{n-2}$, for $n \geq 2$, given $a_1 = 5, a_2 = 3$

Sol Given $a_n = 2a_{n-1} - 2a_{n-2}$, $a_1 = 5, a_2 = 3$

$$\text{--- (1) ---}$$

$$a_n - 2a_{n-1} + 2a_{n-2} = 0.$$

Let us make C as a

$$a_n - 2a_{n-1} + 2a_{n-2} = 0.$$

$$\therefore a_n = 1, a_{n-1} = -2, a_{n-2} = 2.$$

A.E. is $a_n k^2 + a_{n-1}k + a_{n-2} = 0.$

$$k^2 - 2k + 2 = 0.$$

$$k = \frac{-(-2) \pm \sqrt{4-8}}{2}$$

$$k = \frac{2 \pm \sqrt{-4}}{2} = \frac{2 \pm 2i}{2} = 1 \pm i$$

$$k_1 = 1+i, k_2 = 1-i.$$

Comparing with $p \pm iq$, $p=1, q=1.$

$$r = \sqrt{p^2 + q^2} = \sqrt{1+1} = \sqrt{2}.$$

$$\theta = \tan^{-1}\left(\frac{q}{p}\right) = \tan^{-1}\left(\frac{1}{1}\right) = \tan^{-1}(1).$$

$$\theta = \pi/4.$$

$\therefore$ General solution is

$$a_n = r^n [A \cos n\theta + B \sin n\theta]$$

$$a_n = (\sqrt{2})^n [A \cos n\pi/4 + B \sin n\pi/4] \quad \text{--- (2) ---}$$

Given $a_1 = 5, a_2 = 3$

Put $n=1, a_1 = \sqrt{2} [A \cos \pi/4 + B \sin \pi/4]$

3) Solve the recurrence relation

$$a_{n+2} - 3a_{n+1} + 2a_n = 0, \text{ with } a_0 = 1, a_1 = 6.$$

$$a_1 = \sqrt{2} \left[\cancel{\cos} A \cdot \frac{1}{\sqrt{2}} + B \cdot \frac{1}{\sqrt{2}} \right]$$

$$a_1 = A + B \Rightarrow A + B = 5.$$

Put $n=2$,

$$a_2 = \sqrt{2} \left[A \cos \frac{\pi}{2} + B \sin \frac{\pi}{2} \right]$$

$$3 = \sqrt{2} [0 + B(1)]$$

$$\boxed{\frac{3}{\sqrt{2}} = B}$$

$$B = \frac{3}{\sqrt{2}}$$

$$A + B = 5$$

$$A = 5 - B$$

$$\boxed{A = 5 - \frac{3}{\sqrt{2}}} \quad A = \frac{7}{2}$$

Sub (A) & (B) in (2),

$$a_n = (\sqrt{2})^n \left[\left(5 - \frac{3}{\sqrt{2}} \right) \cos \frac{n\pi}{4} + \frac{3}{\sqrt{2}} \sin \frac{n\pi}{4} \right]$$

$$\therefore C_n = (\sqrt{2})^n \left[\left(5 - \frac{3}{\sqrt{2}} \right) \cos \frac{n\pi}{4} + \frac{3}{\sqrt{2}} \sin \frac{n\pi}{4} \right]$$

$$C_n = (\sqrt{2})^n \left[\frac{7}{2} \cos \frac{n\pi}{4} + \frac{3}{2} \sin \frac{n\pi}{4} \right]$$

3) Solve the recurrence relation

$$a_n - 6a_{n-1} + 9a_{n-2} = 0 \text{ for } n \geq 2, \text{ given that } a_0 = 5, a_1 = 12.$$

Sol. given $a_n - 6a_{n-1} + 9a_{n-2} = 0$, $a_0 = 5, a_1 = 12$

①

$$C_n = 1, C_{n-1} = -6, C_{n-2} = 9$$

$$\text{A.E. is } k^2 - 6k + 9 = 0$$

$$k = 3, 3$$

Roots are real and repeated.

$\therefore$ general solution of ① is

$$a_n = (A + Bn) \cdot k^n$$

$$a_n = (A + Bn) 3^n \quad \text{--- ②}$$

Put $n=0$ in ②,

$$a_0 = A \cdot (3)^0$$

$$A = a_0$$

$$\boxed{A = 5}$$

Put $n=1$ in ②,

$$a_1 = (A + B) \cdot 3$$

$$12 = (5 + B) \cdot 3$$

$$4 = 5 + B$$

$$\boxed{B = -1}$$

sub. ① & ② in ②,

$$\boxed{a_n = (5 - n) 3^n}$$

H) Solve the recurrence relation.

~~710-8) c)~~ $C_n \rightarrow 2 \rightarrow a_{n+2} - 3a_{n+1} + 2a_n = 0, a_0 = 1, a_1 = 6$

~~sol~~ Given $a_{n+2} - 3a_{n+1} + 2a_n = 0$. $a_0 = 1, a_1 = 6$.
— (1)

$$C_{n+2} = 1, C_{n+1} = -3, C_n = 2.$$

AE is $C_{n+2} \cdot k^2 + C_{n+1} \cdot k + C_n = 0$,

$$k^2 - 3k + 2 = 0.$$

$$k = \frac{3 \pm \sqrt{9 - 8}}{2} = \frac{3 \pm 1}{2}$$

$$k = 2, 1.$$

$k_1 = 2, k_2 = 1$, real & distinct roots.

$\therefore$ General Solution of (1) is

$$a_n = A(2)^n + B(1)^n.$$

$$a_n = A(2)^n + B. \text{ — (2)}$$

Put $n=0$, in (2), $a_0 = A(2)^0 + B$.

$$1 = A + B \Rightarrow B = 1 - A$$

Put $n=1$, in (2), $a_1 = A(2) + B$

$$6 = 2A + B$$

$$6 = 2A + 1 - A$$

$$6 - 1 = A$$

$$A = 5, B = -4.$$

$\therefore$ Put in (2), we get, $a_n = 5 \cdot (2)^n - 4$

(5) solve the recurrence relation $F_{n+2} = F_{n+1} + F_n$ where
 $n \geq 0$ and $F_0 = 0$, $F_1 = 1$.

Sol. Given $F_{n+2} = F_{n+1} + F_n$, $F_0 = 0$, $F_1 = 1$

$$F_{n+2} - F_{n+1} - F_n = 0. \quad \text{--- (1)}$$

AE is $k^2 - k - 1 = 0$.

$$k = \frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2}, \text{ real \& distinct.}$$

$$k_1 = \frac{1+\sqrt{5}}{2}, \quad k_2 = \frac{1-\sqrt{5}}{2}.$$

General soln of (1) is

$$F_n = A \left(\frac{1+\sqrt{5}}{2} \right)^n + B \left(\frac{1-\sqrt{5}}{2} \right)^n \quad \text{--- (2)}$$

Put $n=0$ in (2),

$$F_0 = A + B.$$

$$0 = A + B \quad \Rightarrow \quad A = -B,$$

Put $n=1$ in (2),

$$F_1 = A \left(\frac{1+\sqrt{5}}{2} \right) + B \left(\frac{1-\sqrt{5}}{2} \right).$$

$$F_1 = -B \left(\frac{1+\sqrt{5}}{2} \right) + B \left(\frac{1-\sqrt{5}}{2} \right)$$

$$1 = B \left(\frac{1-\sqrt{5}}{2} - \left(\frac{1+\sqrt{5}}{2} \right) \right)$$

$$1 = \frac{B}{2} [\cancel{1} - \sqrt{5} - \cancel{1} - \sqrt{5}] \Rightarrow 1 = \frac{B}{2} [-2\sqrt{5}]$$

$$B = -\frac{1}{\sqrt{5}}$$

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$$\therefore A = -B = \frac{1}{\sqrt{5}}$$

Sub. (A) & (B) in (2),

$$F_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$$

$$F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right]$$

$\Rightarrow$

6) Solve the recurrence relation

$$D_n = b D_{n-1} - b^2 D_{n-2}, \quad n \geq 3 \text{ given } D_1 = b > 0$$

and $D_2 = 0$.

Sol

Given $D_n = b D_{n-1} - b^2 D_{n-2}, \quad n \geq 3, \quad D_1 = b > 0, \quad D_2 = 0.$

$$D_n - b D_{n-1} + b^2 D_{n-2} = 0 \quad \text{--- (1)}$$

A.E. is

$$k^2 - b k + b^2 = 0.$$

$$k = \frac{b \pm \sqrt{b^2 - 4b^2}}{2}; \quad \frac{b \pm \sqrt{-3b^2}}{2}$$

$$k = \frac{b \pm i\sqrt{3} b}{2}, \text{ imaginary roots}$$

$$k = \frac{b}{2} \pm \frac{i\sqrt{3}b}{2}.$$

$\therefore$ general solution is

$$D_n = r^n [A \cos n\theta + B \sin n\theta] \quad \text{--- (2)}$$

$$p = \frac{b}{2}, \quad q = \frac{\sqrt{3}b}{2}$$

$$r = \sqrt{p^2 + q^2} = \sqrt{\frac{b^2}{4} + \frac{3b^2}{4}} = b$$

$$\theta = \tan^{-1}\left(\frac{q}{p}\right) = \tan^{-1}\left(\frac{\sqrt{3}b/2}{b/2}\right) = \tan^{-1}\sqrt{3} = \frac{\pi}{3}.$$

$$\textcircled{2} \Rightarrow D_n = b^n \left[A \cos n\frac{\pi}{3} + B \sin n\frac{\pi}{3} \right] \quad \text{--- (3)}$$

$$D_1 = b, \quad D_2 = 0.$$

Put $n=1$ in (3),

$$D_1 = b \left[A \cos \frac{\pi}{3} + B \sin \frac{\pi}{3} \right]$$

$$b = b \left[A \frac{1}{2} + B \frac{\sqrt{3}}{2} \right]$$

$$2 = A + \sqrt{3} B \quad \text{--- (4)}$$

Put $n=2$ in (3),

$$D_2 = b^2 \left[A \cos 2\pi/3 + B \sin 2\pi/3 \right]$$

$$0 = b^2 \left[A \cos 2\pi/3 + B \sin 2\pi/3 \right]$$

$$0 = b^2 \left[A \cos (\pi - \pi/3) + B \sin (\pi - \pi/3) \right]$$

$$0 = -A \cos \pi/3 + B \sin \pi/3$$

$$0 = A \cos 2\pi/3 + B \sin 2\pi/3$$

$$0 = -\frac{1}{2} A + \frac{\sqrt{3}}{2} B$$

$$-A + \sqrt{3} B = 0 \quad \text{--- (5)}$$

solving (4) & (5),

$$A = 1, \quad B = \frac{1}{\sqrt{3}}$$

Put in (3),

$$D_n = b^n \left[\cos n\pi/3 + \frac{1}{\sqrt{3}} \sin n\pi/3 \right]$$

In Calculator

$$\cos\left(\frac{2\pi}{3}\right) = \cos\left(\frac{2 \times 180}{3}\right) = -\frac{1}{2}$$

Radian Degree

mode

Rook Polynomial:

- * Consider a board that represents a part of a chess board.
- * Let n be the no. of squares present in the board.
- * Pawns (Rooks) are placed on the board such that not more than one pawn occupies a square. Then we cannot use more than ' n ' pawns.
- * Two pawns (rooks) placed on a board are said to capture (or take) each other if they are in the same row or same column of the board.
- * Let r_k denote the no. of ways in which ' k ' rooks can be placed on a board such that no two rooks capture each other.
- * If the board is denoted by ' C ', then the rook polynomial is given by $r(C, x) = 1 + r_1 x + r_2 x^2 + \dots + r_n x^n$
 $r_1 = n$, where n is the no. of squares in given board.
 [∵ r_1 denotes no. of ways 1 pawn can be placed on a board. 1st pawn can be placed anywhere on the board. ∴ $r_1 = n$]

$$r(C, x) = 1 + nx + r_2 x^2 + \dots + r_n x^n.$$

- * Rook polynomial is defined for $n \geq 2$.
 If $n=1$, then the board contains only 1 square.
 ∴ $r_2, r_3, \dots$ are zeros.
 ∴ $r(C, x) = 1 + x$.

- * Rook polynomial for $n \times n$ board is given by:

$$r(C_{n \times n}, x) = 1 + \binom{n}{1}^2 x + 2! \times \binom{n}{2}^2 x^2 + 3! \times \binom{n}{3}^2 x^3 + \dots + n! \times \binom{n}{n}^2 x^n$$

Expansion formula:-

In a given board 'C', suppose we choose a particular square and mark it as $\otimes$. Let 'D' be the board obtained from 'C' by deleting the row and column containing the square $\otimes$ and 'E' be the board obtained from 'C' by deleting only the square $\otimes$, then the rook polynomial for the board 'C' is given by

$$r(C, x) = x \cdot r(D, x) + r(E, x).$$

This is known as expansion formula for $r(C, x)$.

Product formula:

Suppose a board is made up of 2 parts C_1 and C_2 where C_1 and C_2 have no square in the same row or column of C. Then the rook polynomial for the board C is given by

$$r(C, x) = r(C_1, x) \times r(C_2, x).$$

This is known as product formula for $r(C, x)$.

Arrangements with forbidden positions:-

Suppose 'm' objects are to be arranged in 'n' places where $n \geq m$. Suppose there are constraints under which some objects cannot occupy certain places. Such places are called the forbidden positions for the said objects.

The no. of ways of carrying out this task is given by the following rule:

$$N = S_0 - S_1 + S_2 - S_3 + \dots + (-1)^n S_n.$$

where $S_0 = n!$, $S_k = (n-k)! \times r_k$, for $k = 1, 2, \dots, n$

where r_k is the coefficient of x^k in the rook polynomial of the board of m rows and n columns where squares represent the forbidden places.

1) Consider the board

1	2
3	4

 Find the rook polynomial

(18)

Sol

Here $n=4$.

The positions of 2 non-capturing rooks are: $(1, 4), (2, 3)$.

$$\therefore r_2 = 2.$$

The board has no positions for more than 2 non-capturing rooks.

$$\therefore r_3 = r_4 = 0$$

$\therefore$ Rook polynomial is $r(C, x) = 1 + 4x + 2x^2$.

2)

	1	2
	3	4
5	6	7

Here $n=7$.

Positions of 2 non-capturing rooks are:

$(1, 4), (1, 5), (1, 7), (2, 3), (2, 5), (2, 6), (3, 5), (3, 7), (4, 5)$
 $(4, 6)$.

$$\therefore r_2 = 10$$

Positions of 3 non-capturing rooks are:

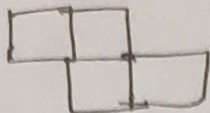
$(1, 4, 5), (2, 3, 5)$

$$\therefore r_3 = 2.$$

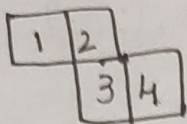
$$r_4 = \dots = r_7 = 0$$

Rook polynomial is $r(C, x) = 1 + 7x + 10x^2 + 2x^3$.

3) Find the rook polynomial the figure.



Sol



Here $n=4$.

No. of 2 non-capturing rooks are $(1,3), (1,4), (2,4)$

$$\therefore r_2 = 3$$

$$r_3 = 0$$

$\therefore$ Rook polynomial $r(C, x) = 1 + nx + r_2x^2$

$$r(C, x) = 1 + 4x + 3x^2$$

4) By using expansion formula, find the rook polynomial for the Board C. (20)

1		2
3	4	5
	6	

C.

1		2
3	4	5
	6	

C

D:

		2
	6	

E:

1		2
	4	5
	6	

Rook Polynomial for D,

$$n=2, r_0=1$$

$$\therefore r(D, x) = 1 + 2x + x^2.$$

Rook Polynomial for E,

$$n=5.$$

The positions for 2 non-capturing rooks are

$$(1,4), (1,5), (1,6), (2,4), (2,6), (5,6).$$

$$\therefore r_2 = 6.$$

Also $r_3 = 1.$ $(1,5,6).$

$$\therefore r(E, x) = 1 + 5x + 6x^2 + x^3$$

Thus By expansion formula,

$$r(C, x) = x r(D, x) + r(E, x)$$

$$\begin{aligned} r(C, x) &= x(1 + 2x + x^2) + (1 + 5x + 6x^2 + x^3) \\ &= 1 + 6x + 8x^2 + 2x^3 \end{aligned}$$

5) Find the rook polynomial for 3×3 board using expansion formula.

Sol

1	2	3
4	5	6
7	8	9

C.

1	3
7	9

D.

1	2	3
4	/	6
7	8	9

E.

For the board D,

$$r_1 = 4, \quad r_2 = 2 \quad (1,9), (3,7).$$

$$r_3 = r_4 = 0$$

$$\therefore r(D, x) = 1 + 4x + 2x^2.$$

For the board E.

$$r_1 = 8.$$

$$r_2 = 14 \quad \left[(1,6), (1,8), (1,9), (2,4), (2,6), (2,7), (2,8), \right. \\ \left. (3,4), (3,7), (3,8), (4,8), (4,9), (6,7), \right. \\ \left. (6,8) \right]$$

$$r_3 = 4 \quad \left[(1,6,8), (2,4,9), (2,6,7), (3,4,8) \right]$$

$$r_4 = \dots = r_8 = 0.$$

$$\therefore r(E, x) = 1 + 8x + 14x^2 + 4x^3.$$

Thus Rook polynomial for 3×3 board is (Using expansion formula)

$$r(C, x) = x r(D, x) + r(E, x)$$

$$= x(1 + 4x + 2x^2) + 1 + 8x + 14x^2 + 4x^3$$

$$r(C, x) = 1 + 9x + 18x^2 + 6x^3$$

Cont. of (6).

(22)

$$\gamma(C, x) = 1 + 11x + 40x^2 + 56x^3 + 28x^4 + 4x^5$$

$$\gamma_1 = 11, \gamma_2 = 40, \gamma_3 = 56, \gamma_4 = 28, \gamma_5 = 4.$$

We have $S_0 = n! = 5! = 120$

$$S_1 = (5-1)! \times \gamma_1 = 4! \times 11 =$$

$$S_2 = (5-2)! \times \gamma_2 = 3! \times 40 =$$

$$S_3 = (5-3)! \times \gamma_3 = 2! \times 56 =$$

$$S_4 = (5-4)! \times \gamma_4 = 1! \times 28 =$$

$$S_5 = (5-5)! \times \gamma_5 = 0! \times 4 = 4.$$

$$\therefore \bar{N} = S_0 - S_1 + S_2 - S_3 + S_4 - S_5$$

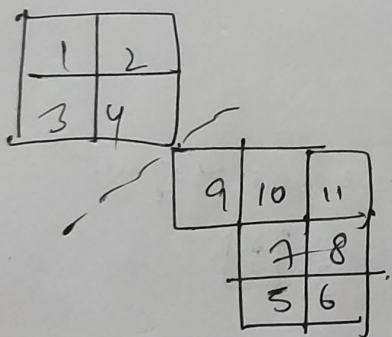
$$\bar{N} = 8.$$

67. Five teachers T_1, T_2, T_3, T_4, T_5 are to be made class teachers for 5 classes. C_1, C_2, C_3, C_4, C_5 , One teacher for each class. T_1 and T_2 do not wish to become the class teacher for C_1 or C_2 , T_3 and T_4 for C_4 or C_5 and T_5 for C_3 or C_4 or C_5 . In how many ways can the teachers be assigned the work, (without displeasing any teacher)

Sol Let us construct a table where rows are ^{teachers} classes and columns are ^{teacher} classes. Shade the forbidden positions

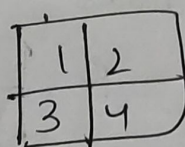
	C_1	C_2	C_3	C_4	C_5
T_1					
T_2					
T_3					
T_4					
T_5					

Let C be the board consisting of shaded region

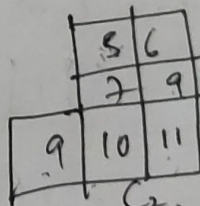


C

This can be divided into 2 distinct disjoint subparts -



C_1



C_2

$$\therefore r(C, x) = r(C_1, x) \times r(C_2, x)$$

$$r(C_1, x): n=4, r_1=2$$

$$r(C_1, x) = 1 + 4x + 2x^2$$

$$r(C_2, x): n=7, r_1=10, r_3=2$$

$$\therefore r(C_2, x) = 1 + 7x + 10x^2 + 2x^3$$

$$\therefore r(C, x) = r(C_1, x) \times r(C_2, x)$$

$$= (1 + 4x + 2x^2) \times (1 + 7x + 10x^2 + 2x^3)$$

$$r(C, x) = 1 + 11x + 40x^2 + 56x^3 + 28x^4 + 4x^5$$

7. An apple, a banana, a mango and an orange are to be distributed among 4 boys B_1, B_2, B_3, B_4 . The boy B_1 and B_2 don't want to have an apple, the boy B_3 doesn't want banana & mango and B_4 refuses orange. In how many ways the distribution can be made so that no boy is displeased.

sol Let us construct a table where rows are Apple, mango, Banana and Orange and columns are B_1, B_2, B_3, B_4 .

Shade the forbidden positions.

	B_1	B_2	B_3	B_4
A				
B				
M				
O				

Let C be the board consisting of shaded positions

-1	2
	3
	4
	5

This can be divided into 3 disjoint subset

$$C_1: \begin{bmatrix} 1 & 2 \end{bmatrix}$$

$$C_2: \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$C_3: \begin{bmatrix} 5 \end{bmatrix}$$

$$\therefore r(C, x) = r(C_1, x) \times r(C_2, x) \times r(C_3, x). \quad \text{--- (1)}$$

$$C_1: n=2, \quad r_2=0 \Rightarrow r(C_1, x) = 1+2x.$$

$$C_2: n=2, \quad r_2=0 \Rightarrow r(C_2, x) = 1+2x.$$

$$C_3: n=1 \Rightarrow r(C_3, x) = 1+x.$$

$$\textcircled{1} \Rightarrow r(C, x) = (1+2x)(1+2x)(1+x).$$

$$r(C, x) = 1 + 5x + 8x^2 + 4x^3$$

Thus for board C ,

$$r_1 = 5, \quad r_2 = 8, \quad r_3 = 4.$$

We have

$$S_0 = 4! = 24$$

(26)

$$S_1 = (4-1)! \times 8_1 = 6 \times 5 = 30$$

$$S_2 = (4-2)! \times 8_2 = 2 \times 8 = 16$$

$$S_3 = (4-3)! \times 8_3 = 1 \times 4 = 4$$

$$\therefore \bar{N} = S_0 - S_1 + S_2 - S_3$$

$$= 24 - 30 + 16 - 4$$

$$\bar{N} = 6$$

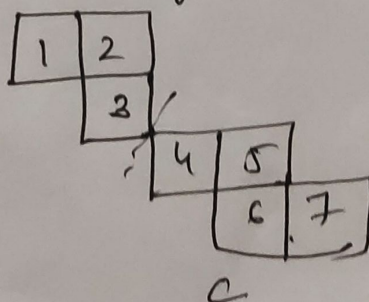
$\therefore$ There are 6 ways of distributing fruits so that no one is displeased.

87. Four persons P_1, P_2, P_3, P_4 who arrives late for a dinner party find that only one chair at each of five tables T_1, T_2, T_3, T_4 and T_5 is vacant. P_1 will not sit at T_1 or T_2 , P_2 will not sit at T_2 , P_3 will not sit at T_3 or T_4 and P_4 will not sit at T_4 or T_5 . Find the number of ways they can occupy vacant chair

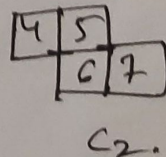
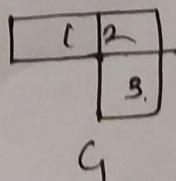
Sol

	T_1	T_2	T_3	T_4	T_5
P_1					
P_2					
P_3					
P_4					

consider C having shaded square



Let C_1 and C_2 be disjoint boards



C_1

C_2

$$r(C_1 x) = r(C_1, x) \times r(C_2, x). \text{ --- (1)}$$

$$C_1: n=3, r_2=1 \Rightarrow r(C_1, x) = 1+3x+x^2$$

$$C_2: n=4, r_2=3 \Rightarrow r(C_2, x) = 1+4x+3x^2$$

$$\textcircled{1} \Rightarrow r(C_1 x) = (1+3x+x^2)(1+4x+3x^2)$$

$$r(C, x) = 1+7x+16x^2+13x^3+3x^4$$

Thus for the board 'C',

$$r_1=7, r_2=16, r_3=13, r_4=3$$

$$\therefore \text{Required no. of ways} = S_0 - S_1 + S_2 - S_3 + S_4$$

$$= 5! - (5-1)! \times r_1 + (5-2)! \times r_2 - (5-3)! \times r_3 + (5-4)! \times r_4$$

$$= 5! - 24 \times 7 + 6 \times 16 - 2 \times 13 + 1 \times 3$$

$$= 120 - 168 + 96 - 26 + 3$$

$$= 25$$

Derangements:

A permutation of ' n ' objects in which none of the objects is in its natural place (original place) is called a derangement.

The no. of possible derangements of ' n ' distinct objects is denoted by d_n and is given by

$$d_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!}$$

$$d_n = n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots - \frac{(-1)^n}{n!} \right] \quad \text{--- (1)}$$

$$\Rightarrow d_1 = 0, d_2 = 1, d_3 = 2, \dots$$

$$* \text{ WKT } e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots$$

$$e^{-1} = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots \quad \begin{matrix} \approx 0.3679 \\ \text{(for } n \geq 7) \end{matrix}$$

$$\therefore d_n = n! \times e^{-1} \quad (\text{for } n \geq 7)$$

Eg: Permutation of ' n ' distinct objects $1, 2, 3, \dots, n$ in which 1 is not in first place, 2 is not in second place, $\dots$, n is not in n^{th} place is a derangement.

1) Find the no. of derangements of 1, 2, 3, 4. List all derangements.

Sol

$$d_n = n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + \frac{(-1)^n}{n!} \right]$$

$$d_4 = 4! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right]$$

$$d_4 = 24 \left[1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} \right]$$

$$d_4 = 9.$$

The derangements are

2	3	4	1	3	4	1	2	4	1	2	3
2	4	1	3	3	4	2	1	4	3	1	2
2	1	4	3	3	1	4	2	4	3	2	1

2) In how many ways can each of 10 people select a left glove and a right glove out of total of 10 pairs of gloves so that no person selects a matching pair of gloves

Sol There are 10 people and 10 pairs of gloves

[The left glove can be given in $10!$ ways] $\angle$

For no person to get matching pair, d_n ways can be done.

$$d_n = n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + \frac{(-1)^n}{n!} \right]$$

$$d_{10} = 10! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} - \frac{1}{7!} + \frac{1}{8!} - \frac{1}{9!} + \frac{1}{10!} \right]$$

$$d_{10} = 10! \times e^{-1} \approx \frac{13,349,600}{916} \quad [10! \times 0.3679]$$

Principles of counting:

$$x \quad |A \cup B| = |A| + |B| - |A \cap B|$$

$$* \quad A \cap \bar{B} = \overline{A \cup B} \quad , \quad |\overline{A \cup B}| = |S| - |A \cup B| \quad \left. \vphantom{A \cap \bar{B}} \right\} |A| = |S| - |A|$$

$$x \quad |\bar{A} \cap \bar{B}| = |\overline{A \cup B}| = |S| - |A \cup B|$$

$$|\overline{A \cup B}| = |S| - |A| - |B| + |A \cap B|$$

— Principle of inclusion - exclusion for 2 sets

Principle of Inclusion - Exclusion for 'n' sets:

Let S be a finite set and $A_1, A_2, \dots, A_n$ be subsets of S ,
then $|A_1 \cup A_2 \cup \dots \cup A_n| = \sum |A_i| - \sum |A_i \cap A_j| + \dots$

$$|A_1 \cup A_2 \cup \dots \cup A_n| = |S| - \frac{\sum |A_i \cap A_j \cap A_k| + \dots + (-1)^{n-1} |A_1 \cap A_2 \cap \dots \cap A_n|}{|A_1 \cup A_2 \cup \dots \cup A_n|}$$

intersection of all the sets

$$|A_1 \cup A_2 \cup \dots \cup A_n| = |S| - \sum |A_i| + \sum |A_i \cap A_j| - \sum |A_i \cap A_j \cap A_k| + \dots + (-1)^{n-1} |A_1 \cap A_2 \cap \dots \cap A_n|$$

* The no. of elements in S that satisfy exactly m of the given n conditions ($0 \leq m \leq n$) is given by

$$E_m = S_m - \binom{m+1}{1} S_{m+1} + \binom{m+2}{2} S_{m+2} - \dots + (-1)^{n-m} \binom{n}{n-m} S_n$$

* The no. of elements in S that satisfy atleast ' m ' of the ' n ' conditions ($0 \leq m \leq n$) is given by

$$E_m = S_m - \binom{m}{m-1} S_{m+1} + \binom{m+1}{m-1} S_{m+2} - \dots + (-1)^{n-m} \binom{n-1}{m-1} S_n$$

* Let $S_0 = |S|$, $S_1 = \sum |A_i|$, $S_2 = \sum |A_i \cap A_j|$,

$$S_3 = \sum |A_i \cap A_j \cap A_k|, \dots, S_n = |A_1 \cap A_2 \cap \dots \cap A_n|$$

then $|A_1 \cup A_2 \cup \dots \cup A_n| = S_0 - S_1 + S_2 - S_3 + \dots + (-1)^n S_n$

1) Among the students in a hostel, 12 students study mathematics (A), 20 study physics (B), 20 study chemistry (C) and 8 study biology (D). There are 5 students for A & B, 7 for A & C, 4 for A & D, 16 for B & C, 4 for B & D, 3 for C & D. There are 3 students for A, B & C, 2 for A, B & D, 2 for B, C & D, 3 for A, C & D. Finally there are 2 who study all of these subjects. Further, there are 71 students who do not study any of these subjects. Find the total no. of students in the hostel.

sol. Given $|A| = 12, |B| = 20, |C| = 20, |D| = 8$.

$$|A \cap B| = 5, |A \cap C| = 7, |A \cap D| = 4, |B \cap C| = 16, |B \cap D| = 4,$$

$$|C \cap D| = 3, |A \cap B \cap C| = 3, |A \cap B \cap D| = 2, |B \cap C \cap D| = 2, |A \cap C \cap D| = 3,$$

$$|A \cap B \cap C \cap D| = 2, |\bar{A} \cap \bar{B} \cap \bar{C} \cap \bar{D}| = 71$$

$$|\bar{A} \cap \bar{B} \cap \bar{C} \cap \bar{D}| = |\overline{A \cup B \cup C \cup D}|$$

$$|\bar{A} \cap \bar{B} \cap \bar{C} \cap \bar{D}| = |S| - |A \cup B \cup C \cup D| \quad \text{--- (1)}$$

$$\begin{aligned}
 |A \cup B \cup C \cup D| &= [|A| + |B| + |C| + |D|] - [|A \cap B| + |A \cap C| + |A \cap D| \\
 &\quad + |B \cap C| + |B \cap D| + |C \cap D|] + [|A \cap B \cap C| + |A \cap B \cap D| + \\
 &\quad + |B \cap C \cap D| + |A \cap C \cap D|] - |A \cap B \cap C \cap D|.
 \end{aligned}$$

$$\begin{aligned}
 |A \cup B \cup C \cup D| &= 12 + 20 + 20 + 8 - 5 - 7 - 4 - 16 - 4 - 3 - 3 + 3 \\
 &\quad + 2 + 2 + 3 + 2
 \end{aligned}$$

$$|A \cup B \cup C \cup D| = 29.$$

$$\therefore |\bar{A} \cap \bar{B} \cap \bar{C} \cap \bar{D}| = |S| - |A \cup B \cup C \cup D|$$

$$71 = |S| - 29 \Rightarrow |S| = 71 + 29.$$

$$\therefore |S| = 100$$

$\therefore$ Total no. of students in hostel is 100.

Q. Determine the no. of positive integers 'n' such that $1 \leq n \leq 100$ and n is not divisible by 2, 3 or 5.

Sol $S = \{1, 2, 3, \dots, 100\}$

Let A be the set of no. that are divisible by 2

$$|A| = \frac{100}{2} = 50.$$

Let B be the set of no. that are divisible by 3.

$$|B| = \left\lfloor \frac{100}{3} \right\rfloor = \lfloor 33.33 \rfloor = 33$$

Let C be the set of no. that are divisible by 5

$$|C| = \frac{100}{5} = 20.$$

If we need to find $|\bar{A} \cap \bar{B} \cap \bar{C}|$.

$$|\bar{A} \cap \bar{B} \cap \bar{C}| = |\overline{A \cup B \cup C}|$$

$|A \cap B|$ = no. of elements divisible by both 2 and 3 (ie by 6).

$$|A \cap B| = \left\lfloor \frac{100}{6} \right\rfloor = \lfloor 16.66 \rfloor = 16$$

$$|A \cap C| = \left\lfloor \frac{100}{10} \right\rfloor = 10$$

$$|B \cap C| = \left\lfloor \frac{100}{15} \right\rfloor = 6.$$

$|A \cap B \cap C|$ = no. of elements divisible by 2, 3 and 5 (ie by 30)

$$|A \cap B \cap C| = \left\lfloor \frac{100}{30} \right\rfloor = 3.$$

$$\begin{aligned}
 |\overline{A \cap B \cap C}| &= |\overline{A \cup B \cup C}| \\
 &= |S| - |A \cup B \cup C| \\
 &= |S| - (|A| + |B| + |C|) + (|A \cap B| + |B \cap C| + |A \cap C|) \\
 &\quad - |A \cap B \cap C|
 \end{aligned}$$

$$= 100 - (50 + 33 + 20) + (16 + 10 + 6) - 3$$

$$= 26.$$

No. of elements that are not divisible by 2, 3 or 5 is 26.

3) In how many ways 5 no. of a's, 4 no. of b's and 3 no. of c's can be arranged so that all identical letters are not in a single block?

Sol. There are a total of 12 letters in which there are 5 a's, 4 b's and 3 c's.

Let S be the set of all arrangements of 12 letters.

$$\therefore |S| = \frac{12!}{5!4!3!} = 27,720.$$

Let A_1 be the set of all arrangements in which all a's are in a single block.

$$\text{ie } |A_1| = \frac{8!}{4!3!1!} = 280$$

Consider all a's as 1 block.

aaaaa b b b b c c c
1 block

Let A_2 be the no. of ways of arranging in which 4 b's are in a single block.

$$\text{ie } |A_2| = \frac{9!}{5!3!1!} = 504.$$

Let A_3 be the set of all arrangements in which all 3 c's are in a single block.

$$|A_3| = \frac{10!}{5!4!1!} = 1260.$$

Let $|A_1 \cap A_2|$ be no. of ways of arranging in which all 5 a's and 4 b's are together.

$$|A_1 \cap A_2| = \frac{5!}{1!1!3!} = 20$$

Considering all a's and b's as 1 unit

(aaaaa)(bbbb)ccc

Similarly $|A_2 \cap A_3| = \frac{7!}{5!1!1!} = 42$

aaaaa(bbbbb)ccc

and $|A_1 \cap A_3| = \frac{6!}{1!4!1!} = 30$

(aaaaa)bbbb(ccc)

(aaaaa)(bbbb)(ccc)

and $|A_1 \cap A_2 \cap A_3| = 3! = 6$

$$\begin{aligned} \therefore |\overline{A_1} \cap \overline{A_2} \cap \overline{A_3}| &= |S| - [|\overline{A_1 \cup A_2 \cup A_3}|] \\ &= |S| - |A_1 \cup A_2 \cup A_3| \\ &= |S| - [|A_1| + |A_2| + |A_3|] + [|A_1 \cap A_2| + |A_2 \cap A_3| + |A_1 \cap A_3|] \\ &\quad - |A_1 \cap A_2 \cap A_3| \end{aligned}$$

$$= 27,720 - [260 + 504 + 1260] + [20 + 42 + 30] - 6$$

$$= \underline{\underline{25,762}}$$

48. In how many ways can 26 letters of English alphabet be permuted so that none of the patterns CAR, DOG, PUN or BYTE occurs?

Sol. $|S| = 26!$

Let A_1, A_2, A_3, A_4 be sets of permutations containing the patterns CAR, DOG, PUN and BYTE respectively.

$$|A_1| = 24!$$

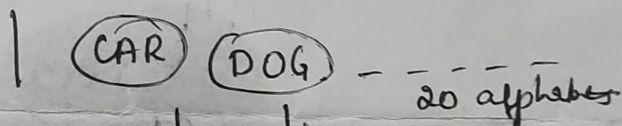
$$|A_2| = 24!$$

$$|A_3| = 24!$$

$$|A_4| = 23!$$

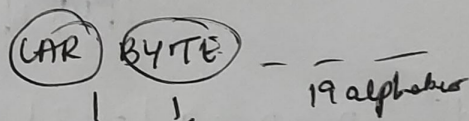
$A_1 \rightarrow$ treat CAR as single unit and other 23 alphabets separately

$$|A_1 \cap A_2| = 22!$$



$$|A_1 \cap A_3| = 22!$$

$$|A_1 \cap A_4| = 21!$$

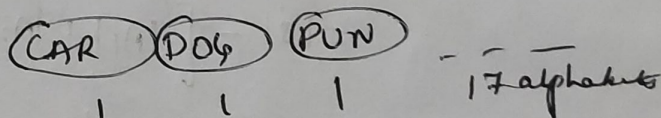


$$|A_2 \cap A_3| = 22!$$

$$|A_2 \cap A_4| = 21!$$

$$|A_3 \cap A_4| = 21!$$

$$|A_1 \cap A_2 \cap A_3| = 20!$$

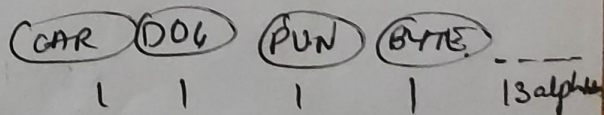


$$|A_1 \cap A_2 \cap A_4| = 19!$$

$$|A_1 \cap A_3 \cap A_4| = 19!$$

$$|A_2 \cap A_3 \cap A_4| = 19!$$

$$|A_1 \cap A_2 \cap A_3 \cap A_4| = 17!$$



$\therefore$ No. of ways in which no pattern occurs is

$$|\overline{A_1} \cap \overline{A_2} \cap \overline{A_3} \cap \overline{A_4}| = |\overline{A_1 \cup A_2 \cup A_3 \cup A_4}| = |S| - |S_1| + |S_2| - |S_3| + |S_4|$$

$$\therefore \text{Ans} = 4.014 \times 10^{26} = 26! - (3 \times 24! + 3 \times 23!) + (3 \times 22! + 3 \times 21!) - (20! + 3 \times 19!) + 17!$$

5% In how many ways one can arrange the letters of the word 36
 CORRESPONDENTS so that there are i) exactly 2 pairs of
 consecutive identical letters? ii) at least 3 pairs of
 consecutive identical letters? (iii) no pair of consecutive identical
 letters?

Sol

CORRESPONDENTS has 1 C, 2 O, 2 R, 2 E, 2 S,
 1 P, 2 N, 1 D, 1 T.

$$\therefore |S| = \frac{14!}{2!2!2!2!2!}$$

Let A_1, A_2, A_3, A_4, A_5 be the sets in which there
 are 2 consecutive O's, R's, E's, S's, N's respectively.

$$\text{Then, } |A_1| = \frac{13!}{(2!)^4} = |A_2| = |A_3| = |A_4| = |A_5|$$

$$|A_i \cap A_j| = \frac{12!}{(2!)^3} \quad \begin{array}{c} \text{OO} \\ \text{---} \end{array} \quad \begin{array}{c} \text{---} \\ \text{---} \end{array} \quad \begin{array}{c} \text{---} \\ \text{---} \end{array} \quad \begin{array}{c} \text{---} \\ \text{---} \end{array} \quad \begin{array}{c} \text{---} \\ \text{---} \end{array}$$

$$\begin{array}{c} \text{OO} \quad \text{RR} \\ \text{---} \quad \text{---} \end{array} \quad \begin{array}{c} \text{---} \\ \text{---} \end{array} \quad \begin{array}{c} \text{---} \\ \text{---} \end{array} \quad \begin{array}{c} \text{---} \\ \text{---} \end{array} \quad \begin{array}{c} \text{---} \\ \text{---} \end{array}$$

$$|A_i \cap A_j \cap A_k| = \frac{11!}{(2!)^2} \quad \begin{array}{c} \text{OO} \quad \text{RR} \quad \text{EE} \\ \text{---} \quad \text{---} \quad \text{---} \end{array} \quad \begin{array}{c} \text{---} \\ \text{---} \end{array} \quad \begin{array}{c} \text{---} \\ \text{---} \end{array}$$

$$|A_i \cap A_j \cap A_k \cap A_l| = \frac{10!}{2!} \quad \begin{array}{c} \text{OO} \quad \text{RR} \quad \text{EE} \quad \text{SS} \\ \text{---} \quad \text{---} \quad \text{---} \quad \text{---} \end{array} \quad \begin{array}{c} \text{---} \\ \text{---} \end{array}$$

$$|A_i \cap A_j \cap A_k \cap A_l \cap A_m| = 9! \quad \begin{array}{c} \text{OO} \quad \text{RR} \quad \text{EE} \quad \text{SS} \quad \text{NN} \\ \text{---} \quad \text{---} \quad \text{---} \quad \text{---} \quad \text{---} \end{array} \quad \begin{array}{c} \text{---} \\ \text{---} \end{array}$$

$$\therefore |S| = \frac{14!}{(2!)^5}$$

$$|S_1| = {}^5C_1 \times \frac{13!}{(2!)^4}$$

$$|S_2| = {}^5C_2 \times \frac{12!}{(2!)^3} \quad |S_4| = {}^5C_4 \times \frac{10!}{2!}$$

$$|S_3| = {}^5C_3 \times \frac{11!}{(2!)^2} \quad |S_5| = {}^5C_5 \times 9!$$

$$(i) \quad |\overline{A_1} \cap \overline{A_2} \cap \overline{A_3} \cap \overline{A_4} \cap \overline{A_5}| = |\overline{A_1 \cup A_2 \cup \dots \cup A_5}|$$

$$= |S| - S_1 + S_2 - S_3 + S_4 - S_5$$

=

$$= 128,60,46,720$$

(i) Exactly 2 pairs of consecutive even letters

$$(ii) \quad E_m = S_m - \binom{m+1}{1} S_{m+1} + \binom{m+2}{2} S_{m+2} - \dots$$

$$E_2 = S_2 - \binom{3}{1} S_3 + \binom{4}{2} S_4 - \binom{5}{3} S_5$$

$$E_2 = \frac{10 \times 12!}{(2!)^3} - 3 \times \frac{10 \times 11!}{(2!)^2} + 4 \times \frac{5 \times 10!}{2!} - 5 \times 9!$$

$$E_2 = 350,179,200$$

~~(ii)~~ No pair of consecutive even letters.

Atleast 3 pairs

$$(iii) \quad L_m = S_m - \binom{m}{m-1} S_{m+1} + \binom{m+1}{m-1} S_{m+2} - \dots$$

$$L_3 = S_3 - \binom{3}{2} S_4 + \binom{4}{2} S_5$$

$$L_3 = \frac{10 \times 11!}{(2!)^2} - 3 \times \frac{5 \times 10!}{2!} + 6 \times 9!$$

$$L_3 = 7,47,53,280$$